Fiscal Year:	FY 2006	**Task Last Updated:**	FY 04/03/2006
PI Name:	Zhao, Yongliang Ph.D.		
Project Title:	Functional Role Of The Betaig-H3 Gene In High-Energy Heavy Ions-Induced Carcinogenesis		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation Biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	10032	**Congressional District:**	15
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	02-OBPR-02
Start Date:	07/01/2003	**End Date:**	06/30/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Changed Division and Discipline/Program to HRP as of FY2006, per program changes then, per JSC/A. Chu-ARC (jvp 4/2009) NOTE: Received NCE to 06/30/2008 per A. Chu/ARC; previously had 6/30/2007 end date (6/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Shao, Genze (Columbia University) Berenguer, Jessica (Columbia University)		
Grant/Contract No.:	NAG2-1637		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Understanding the cellular and molecular mechanism of radiation carcinogenesis induced by high-energy heavy ions is important for both space radiation protection as well as design of interventional approaches. Our previous data demonstrate that loss of expression of Betaig-h3 gene is causally linked to tumorigenic process in high-LET, alpha particle-irradiated human papillomavirus-immortalized human bronchial epithelial (BEP2D) cells. Furthermore, loss of Betaig-h3 expression is common in many human tumor cell lines and reintroduction of this gene into human lung tumor cells completely abrogates their tumorigenicity, suggesting that Betaig-h3 gene may function as a tumor suppressor in a variety of human tumors. The newly established telomerase (hTERT)-immortalized human bronchial epithelial (hTERT-SAEC) cell model that has normal p53 and p16 functions represents an attractive alternative to further ascertain the mechanism involved in radiation carcinogenesis. In this study, we propose to examine the functional role of Betaig-h3 gene in high-energy heavy ions-induced carcinogenesis using the hTERT-SAEC cells and Betaig-h3 knock-out mice. The proposal has 3 main aims: I) To establish the tumorigenic cell model by irradiating hTERT-SAEC cells using beams of charged heavy ions; II) To demonstrate that loss of Betaig-h3 expression is causally related to the malignant transformation process; and III) To generate mouse embryonic stem cells and knock-out mice with disrupted Betaig-h3 gene. Using the Betaig-h3 null mice model, we will address the following questions: (1) whether mice with disruption of this gene predispose to cancer; (2) Is the incidence of transformation induced by heavy ions higher in Betaig-h3 null embryo fibroblast cells than in wild type cells? (3) Does loss of Betaig-h3 gene in embryo fibroblast cells affect radiation sensitivity and apoptosis induced by high-energy heavy ions? I) To establish the tumorigenic cell model by irradiating hTERT-SAEC cells using beams of charged heavy ions; II) To demonstrate that loss of Betaig-h3 expression is causally related to the malignant transformation process; and III) To generate mouse embryonic stem cells and knock-out mice with disrupted Betaig-h3 gene. The data generated from the proposed studies will provide important insights into the role of Betaig-h3 gene in heavy ion-indued carcinogenesis
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Cancer induction by space radiation is a major concern for manned space exploration. Among the various particulate components of ionizing radiation in space, high energy (HZE) heavy ions have been of special concern to radiobiologists and NASA administrators mainly because they contribute about 1% of the particle radiation field in frequency whereas they produce a considerable amount of the whole dose-equivalent because of their large energy deposition and high LET. For the acute effects like cell inactivation are of not so direct interest, late effects including genetic changes and cancer formation are more important for risk estimation. However, for this biological end-point there is no direct human evidence available. The research from this funding project will provide the important information ragarding the malignant transformation and molecular alterations induced by heavy ions.
Task Progress:	1. Human hTERT-immortalized SAEC cells has been irradiated by a single dose of 60 cGy of 56Fe ions at the passages of 20 and 40 in two rounds of experiments in Brook Haven National Laboratory. Following 5 months of continuous culture, the irradiated cells were injected into nude mice subcutaneously and formed progressively growing tumors at 4 months after injection. Two independent tumor cell lines have been extablished from the tumor nodules. 2. We have demonstrated that CpG island hypermethylation of Betaig-h3 promoter is correlated with the gene silencing in different human tumor cell lines including lung, prostate and kidney. 3. Homozygous mice with disruption of Betaig-h3 gene have been shown to have an increased incidence of spontaneous tumor growth.
Bibliography Type:	Description: (Last Updated: 09/29/2008)
Abstracts for Journals and Proceedings	Shao G, Piao CQ, Hei TK, Zhao YL. "Malignant transformation of human small airway epithelial cells induced by heavy ions. " 4th International Workshop on Space Radiation Research, 17th Annual NASA Space Radiation Health Investigators' Workshop, Moscow-St. Petersburg, 5-9 June 2006. 4th International Workshop on Space Radiation Research, 17th Annual NASA Space Radiation Health Investigators' Workshop, 5-9 June 2006. , Jun-2006
Articles in Peer-reviewed Journals	Zhao Y, El-Gabry M, Hei TK. "Loss of Betaig-h3 protein is frequent in primary lung carcinoma and related to tumorigenic phenotype in lung cancer cells." Mol Carcinog. 2006 Feb;45(2):84-92. PMID: 16329146 , Feb-2006
Articles in Peer-reviewed Journals	Shao G, Berenguer J, Borczuk AC, Powell CA, Hei TK, Zhao Y. "Epigenetic inactivation of Betaig-h3 gene in human cancer cells." Cancer Research 66(9), in press, 2006. , May-2006